



Welcome to CSC 1202 (2026)

- Full-Stack Development
 - Django + React
 - Workflow-Based System
 - Coding-Focused Assessment



What This Course Is About

- Designing real systems
 - Writing backend logic
 - Enforcing rules & constraints
 - Deploying working software



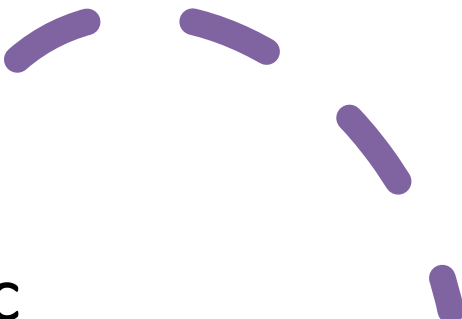
What You Will Build: ILES

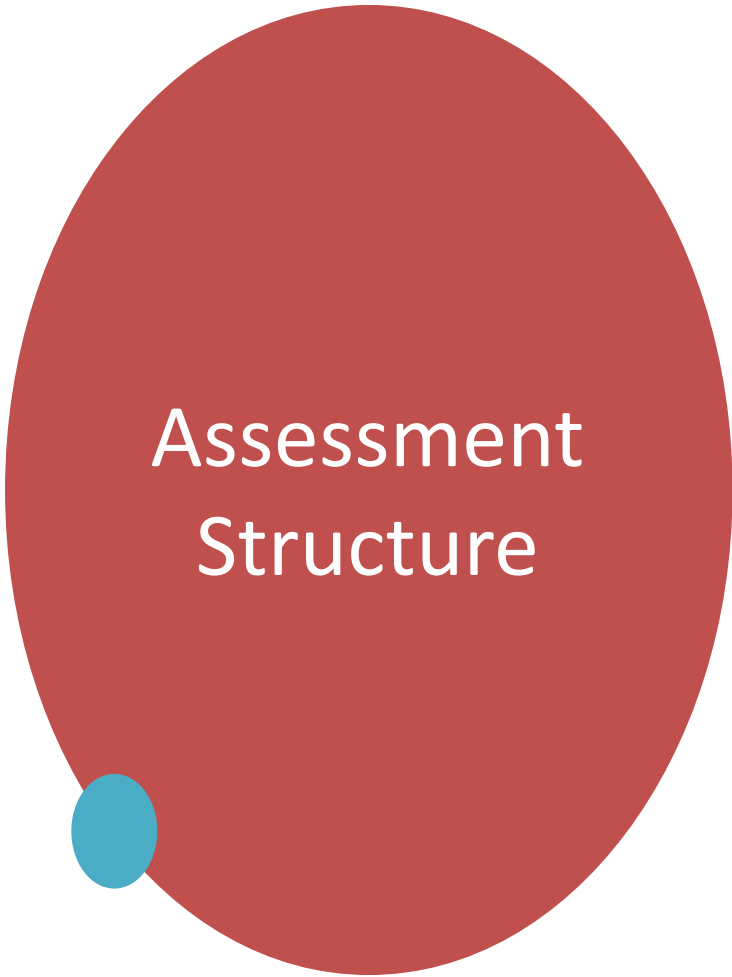
- Internship placements
- Weekly activity logs
- Supervisor reviews
- Academic evaluation
- Weighted scoring
- Dashboards



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Why This Project Is Different

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- Not just CRUD
 - Workflow logic
 - Deadline enforcement
 - Role-based permissions
 - Business rule validation

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Assessment Structure

- 
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- Continuous Assessment – 40%
 - Programming Quizzes – 10%
 - Practical Coding Exam – 40%
 - Final Technical Defense – 10%



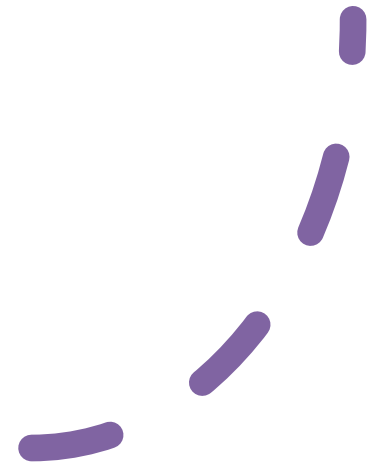
What the Coding Exam Tests

- Writing Django models
 - Serializer validation logic
 - State transition enforcement
 - Fixing buggy code
 - Implementing scoring logic
- 



Thinking Like a Software Engineer

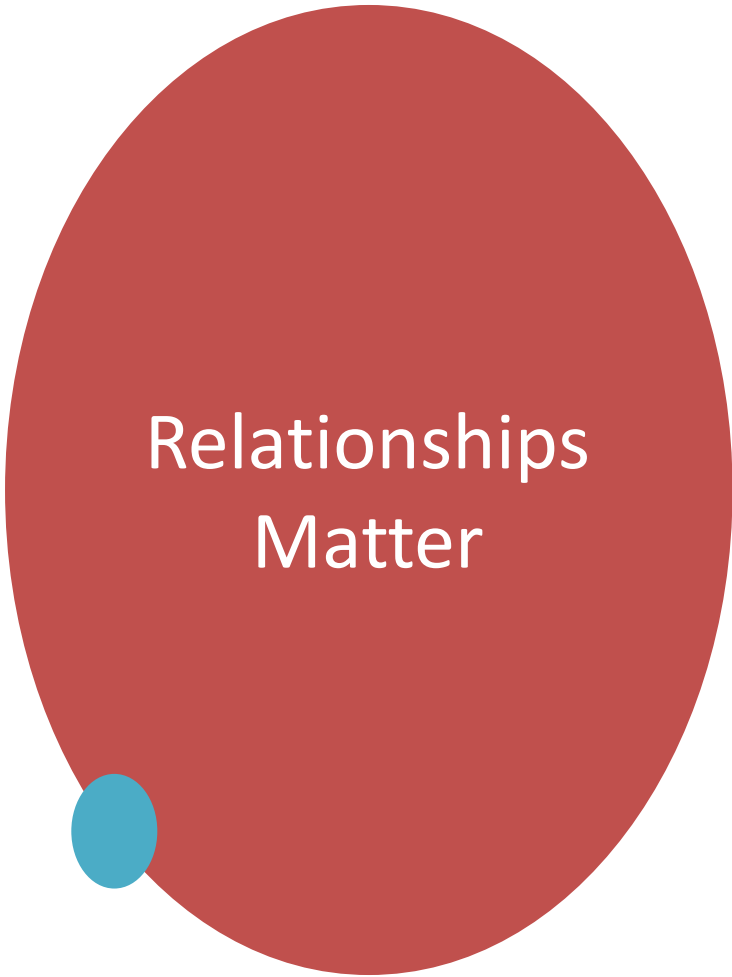
- Software = Data + Rules + States + Behavior





Core Entities in ILES

- User
 - InternshipPlacement
 - WeeklyLog
 - EvaluationCriteria
 - Evaluation
- 

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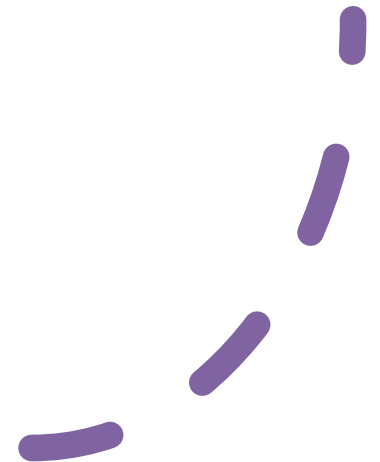
Relationships Matter

- One User → Many WeeklyLogs
 - One WeeklyLog → One Evaluation
 - One User → One Placement
- 
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Workflow States

- Draft
- Submitted
- Reviewed
- Approved





Valid Transitions

- Draft → Submitted
- Submitted → Reviewed
- Reviewed → Approved
- Reviewed → Draft (revision)





Deadline Enforcement

- No submission after deadline
- No editing after approval
- No duplicate evaluation





Weighted Score Example

- Technical Skills – 40%
 - Communication – 30%
 - Professionalism – 30%
 - Final Score = $T*0.4 + C*0.3 + P*0.3$
- 



Why Business Logic Matters

- Logic must live in backend
 - Frontend validation is not enough
 - Rules must be enforced centrally
- 



SDLC Overview

- Planning
 - Analysis
 - Design
 - Implementation
 - Testing
 - Deployment
- 



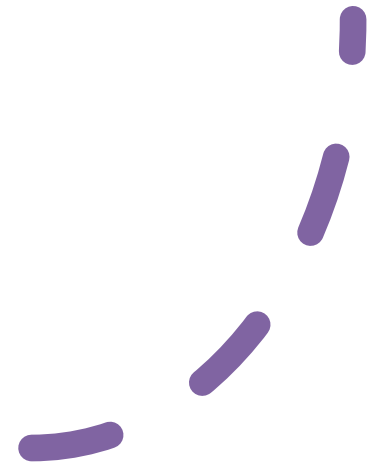
SDLC Applied to ILES

- Planning → Define rules
 - Analysis → Identify workflow
 - Design → ERD & states
 - Implementation → Django + React
 - Testing → Unit tests
 - Deployment → Cloud
- 



Agile in This Course

- Short iterations
- Working software early
- Continuous feedback
- Refactoring





XP Practices We Use

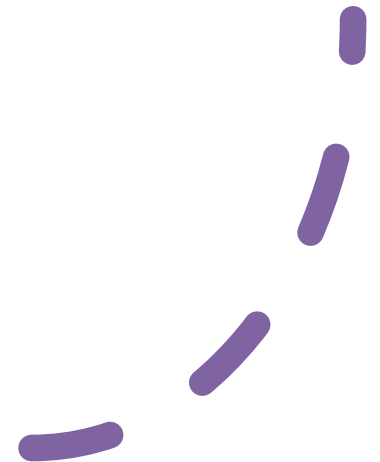
- User stories
- Small increments
- Continuous integration
- Testing mindset





How Coding Skills Develop

- Practice-based
- Cumulative learning
- Pattern recognition
- Problem-solving





Stages of Learning

- Copy examples
 - Modify examples
 - Build small features
 - Build full systems
 - Refactor & optimize
- 



Effective Practice Strategy

- Code daily
 - Debug yourself first
 - Rewrite from memory
 - Build independently
- 



How to Read Code

- What data is used?
 - What state changes?
 - What rule is enforced?
 - What edge case is missing?
- 



Think Before You Code

- Identify entities
 - Define relationships
 - List rules
 - Identify edge cases
- 



AI Is a Tool

- Generates examples
 - Explains syntax
 - Suggests improvements
 - Helps debug
- 



AI Cannot Replace Thinking

- Cannot sit your exam
 - Cannot defend your code
 - Cannot think critically for you
- 



Smart AI Usage Strategy

- Try yourself first
 - Ask for hints
 - Compare solutions
 - Rewrite manually
 - Test independently
- 



How to get a Retake

- Copy full features
- Paste without reading
- Skip debugging
- Never test alone





Academic Integrity

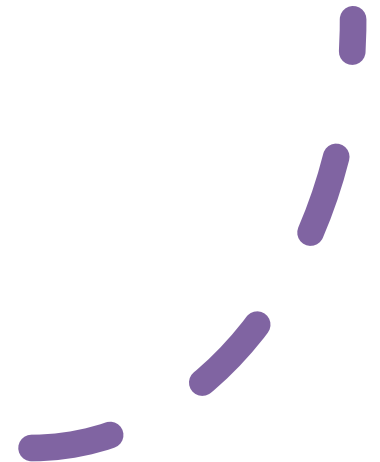
- Coding quizzes
- Practical exam
- Live defense
- Understanding required





What Good Code Looks Like Here

- Clean models
- Valid transitions
- Proper validation
- Tests written
- Clear Git history





Common Failure Patterns

- No validation
- Hardcoded values
- Ignoring edge cases
- Copy-paste coding





Before Next Week

- Install Python & Node
- Install PostgreSQL
- Setup GitHub
- Form teams
- Draft user stories





Final Thought

- This course will stretch you
- It will improve your engineering thinking
- Commit early. Practice daily.

